

Three-stage analysis workflow

Stage 1

Boundary-aware candidate sampling

§3.1

Per-city AOI = ADM2 prefecture polygon
6 cities · taken whole · no urban mask

ESA WorldCover 2021 → 3 classes
stratified sample 5,000 / class → 15,000 / city

Sentinel-2 2021 composite
9 spectral features (6 bands + NDVI, NDBI, MNDWI)
point value + 3×3 mean & SD

BAMS score
RF margin $m = p_1 - p_2 + \text{BoundaryScore}$
 $\text{BoundaryScore} = \sum \sigma_{3 \times 3}(B2, B3, B4, B8) \rightarrow \text{pick 150 / city}$

the random forest is a proposal mechanism only — it assigns no class labels

900 boundary points · independent expert photo-interpretation · QC: 100-pt $\kappa = 0.89$ (92.7 %)

Stage 2

Multi-reference agreement diagnostic

core

§3.2

Does consulting a 2nd / 3rd product (Dynamic World, Esri) arbitrate truth at the urban boundary?

compare expert vs WorldCover · Dynamic World · Esri (n = 900)

(a) Four-way agreement

“how much”

T1

(b) Confidence stratification

annotator noise?

T2

(c) Direction of WC error

“which way”

T3

(d) Physical scene stratification

“where”

T4

(e) Independence & arbitration test

excess agreement · error dependence (Yule Q, κ) · arbitration value · confounds

Tables S2-S3 · Fig 3

Stage 3

LOCO correction operator

supporting

§3.3

leave-one-city-out: fit expert corrections on 5 cities → apply to the held-out 6th (no labels of its own)

the operator overwrites the WorldCover label only where it is confident (gate τ)

Phase 1
operator transfer

Phase 2
downstream map correction

Phase 3
multi-reference referee

before / after correction maps: Fig 4 (main) · Fig S1-S4

Evaluation protocol & leakage controls §3.4

fixed downstream classifier · spatial-block partitioning · target-city isolation · block-bootstrap CIs

All agreement / disagreement rates are conditional on the boundary-candidate population;
they are not map-wide accuracies of any product.